

ARTICLE REVIEW

Modelling Armed Conflict Risk Under Climate Change with Machine Learning and Time-Series Data

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What Does the Paper Ask?

Can climate variability (temperature & precipitation deviations) predict the risk of armed conflict globally, using spatial machine learning on grid-cell data?

Why it matters

91,000+ conflict events globally (2000–2015), 654,000 deaths. Understanding drivers is essential for early warning.

The debate

Climate–conflict link is controversial. Some studies find strong effects, others find none. This paper uses ML to capture non-linear patterns.

The gap

Most prior work uses linear models or country-level data. This paper works at a fine spatial grid ($0.5^\circ \times 0.5^\circ$) with Random Forest.

Data & Spatial Framework

Spatial Unit: PRIO-GRID

- $0.5^{\circ} \times 0.5^{\circ}$ grid cells covering the globe (~55 km at equator)
- Same logic as our course: spatial data on regular grids (Session 2 — grids, rasters, polygons)
- Avoids MAUP issues of admin boundaries (Session 2 — scale vs zone effects)
- Time period: 2000–2015 (annual)

Data Sources

Conflict (Y)

UCDP-GED events → binary incidence / onset per grid-year

Climate

CRU TS — monthly temperature & precipitation → annual deviations from long-term means

Socio-economic

GDP per capita, population, ethnic fractionalization, democracy index (V-Dem)

Geography

Terrain ruggedness, distance to border, land area, latitude

Methodology: Random Forest + Spatial Features

1

Feature Engineering

69 covariates grouped into background context (stable) and climate deviations (dynamic)

2

Random Forest

Classification RF predicting binary conflict incidence / onset per grid-cell-year

3

Variable Importance

Permutation importance (%IncMSE) — same as in Class 5: which features drive predictions?

4

Partial Dependence

Marginal effect of each climate variable — non-linear patterns (Class 5: partialPlot)

Connection to Our Course

Session 2 — Spatial Data Integration

The paper integrates grid data (PRIO-GRID), point data (UCDP conflict events), raster data (CRU climate), and polygon data (country boundaries) into a unified grid framework — the exact workflow from Session 2.

Session 3 — Spatial Weight Matrices & Spatial Lags

Neighbourhood conflict (spatial lag of Y) is one of the strongest predictors. The paper uses spatial diffusion features — how conflict in nearby cells affects the focal cell — directly implementing Wy from Session 3.

Class 5 — Random Forest with Spatial Variables

The core method is RF with spatial features as predictors (distance to border, terrain, latitude). Variable importance (%IncMSE) and partial dependence plots are used exactly as in Class 5.

Session 4 — DEGURBA / Urbanisation

Population density and urbanisation levels feature as key background context variables — paralleling DEGURBA's role as a spatial feature in our models.

Key Findings

~90%

of conflict risk explained by
stable background context

3–20%

additional risk from
climate deviations

AUC 0.8+

strong out-of-sample
predictive performance

What Drives Conflict Risk?

- Neighbourhood conflict history is the #1 predictor — strong spatial autocorrelation (Session 3: Wy spatial lag)
- Background context (GDP, population, ethnic diversity, terrain) explains most of the variance
- Positive temperature deviations → increased conflict risk (non-linear, captured by RF partial plots)
- Precipitation extremes (both droughts and floods) → elevated risk, consistent with resource scarcity theory

Strengths & Limitations

Strengths

- Global coverage at fine spatial resolution (0.5° grid) — avoids country-level aggregation bias
- Random Forest captures non-linear and interaction effects that linear models miss
- 69 covariates with rigorous variable importance analysis (connects to Class 5 methodology)
- Published in Nature Communications — high methodological standard

Limitations

- No explicit spatial weight matrix W in the model — spatial lag is a feature, not part of the algorithm (cf. Spatial RF from Class 5)
- Binary Y (conflict yes/no) — loses intensity information
- Correlation $\neq$ causation: RF identifies patterns but does not prove causal mechanisms
- No Geographically Weighted RF — assumes global relationships (Class 7: GW-RF could improve local prediction)

Takeaways for Discussion

Spatial autocorrelation dominates

Conflict yesterday in a neighbouring cell is the strongest predictor of conflict today. This is exactly the motivation behind spatial weight matrices (Session 3) — "what happens around me matters."

RF works well for spatial prediction

Random Forest with spatial features achieves strong predictive power ($AUC > 0.8$). As we saw in Class 5, including spatial variables (distances, densities) boosts model performance substantially.

Climate is a risk multiplier, not a root cause

Temperature and precipitation deviations add 3–20% to conflict risk on top of existing socio-economic vulnerabilities — not a direct trigger.

Room for improvement with course methods

A natural extension would be Geographically Weighted RF (Class 7) to capture local variation in climate–conflict relationships across regions.

Thank you — questions welcome!